

IBM SkillsBuild for University Engagements

AICTE-2026

Problem Statements on Agentic AI

Problem Statement No. 5 – Travel Planner Agent

The Challenge

A Travel Planner Agent is an AI-powered assistant that helps users plan trips efficiently and intelligently. It uses real-time data to suggest destinations, build itineraries, and recommend transport and accommodation options. By understanding user preferences, budgets, and constraints, it tailors personalized travel plans. Integrated with maps, weather updates, and local guides, it ensures a smooth travel experience. The agent can also manage bookings, alert users to changes, and optimize schedules on the go. This smart assistant transforms complex travel planning into a seamless, enjoyable process.

Technology

Use of IBM Cloud Lite services / IBM Granite is mandatory.

My Solution – AI Travel Planner Agent

Project Overview

I developed an AI-powered Travel Planner Agent using IBM Granite 3-8B Instruct model through WatsonX Orchestrate and Streamlit. The agent helps users plan complete trips by generating day-wise itineraries, recommending hotels and restaurants, suggesting transport options, and managing budgets based on user preferences.

Technologies Used

Technology	Purpose
IBM Granite 3-8B Instruct	Core LLM for travel plan generation
WatsonX Orchestrate	Agentic AI platform for building the Travel Agent
IBM Cloud Lite	Cloud infrastructure and API management
Streamlit	Interactive web UI for user interaction
Python + Requests	Backend API communication
IBM IAM Authentication	Secure token-based API access

Key Features

- Quick Trip Planner – Enter destination, days, budget and interests for instant itinerary generation

- Chat Interface – Conversational AI to ask any travel question naturally
 - Day-wise Itinerary – Structured morning, afternoon and evening activity plans
 - Budget Management – Low/Medium/High budget categories in Indian Rupees
 - Hotel and Transport – Curated recommendations based on budget
 - Food and Culture – Local restaurant suggestions and cultural insights
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